MILITARY SPECIFICATION SHEET

ELECTRON TUBE NEGATIVE GRID (MICROWAVE)

Ⓑ 1/ TYPE 8538B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1 .

DESCRIPTION: Triode, planar, ceramic and metal construction
Ⓑ See figures 1 and 1A
Mounting position: Any
Weight: 0.7 ounce (20 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F GHz	Ef V (Notes 1 & 11)	Eb kVdc	epy kv	Ec Vdc (Note 3)	ib a (Note 2)	Ib mAdc	ic a	tp µs	Du ---
Maximum:										
Anode pulsed osc or amp:	3.0	6.3 ±5%	---	10	-300	5.0	---	2.5	6.0	0.0033
Grid pulsed osc or amp:	3.0	6.3 ±5%	8.0	---	-300	5.0	---	2.5	6.0	0.0033
TEST CONDITIONS:	---	6.3	1.0	---	Adj	---	100	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Pp W	Pg W	tk sec (min)	TE °C (Note 4)	T (Anode shank) °C (Note 4)	Barometric pressure, reduced mmHg (Note 5)	Cooling --- (Note 4)
Maximum:							
Anode pulsed osc or amp:	100	1.5	60	250	250	35	Conduction and convection
Grid pulsed osc or amp:	100	1.5	60	250	250	35	Conduction and convection
TEST CONDITIONS:	---	---	300	---	---	---	Note 6

GENERAL:

Qualification - Required (see note 19)

1/ See note 17

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	7							
1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	R	50	---	Meg
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec	-2.0	-7.0	Vdc
1266	Total grid current	-		0.65	II	Ic	---	-8.0	μAdc
1231	Pulsing emission	-	(eb = ec)/is = 10.0 Ω; tp = 1 to 3 μs; prr = 600 (max)	0.65	II	etd	---	180	v
1301	Heater current	-		0.65	II	If	1.2	1.4	A
---	High-voltage holdoff	9	Ebb = 10 kVdc; Ec = -150 Vdc (max); 10 K ohms (max) anode circuit resistance	0.65	II	Ib	---	1.0	mAdc
	Quality conformance inspection, part 2								
1261	Electrode voltage (2) (grid)	-	Eb = 1.0 kVdc; Ec/Ib = 1.0 mAdc	---	---	Eco	---	-16	Vdc
1331	Direct-interelectrode capacitance	8	No voltages (ground cathode connection); use fixture in accordance with DESC Drawing No. D 72000	---	---	Cin Cgp Cout	8.5 1.30 ---	11.0 1.55 0.06	pF pF pF
---	Power gain (pulse)	16	Grounded grid; Ef = 5.8 V; pd = 300 w (peak); F = 1030 ±5 MHz; Ec = -70 Vdc (min); Eb = 5.0 kVdc; tp = 0.5 to 2.0 μs; Du = 0.001 (max)	---	---	po	2.2	---	dB
	Quality conformance inspection, part 3								
---	Life test	10	Group C; Ef = 6.3 ±0.1 V; heater standby; t = 500 hours	---	---	---	---	---	---
---	Life-test end point	10		---	---	Δib ΔCgp	--- ---	25 0.15	% pF

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Sweep-frequency vibration	8, 12, 13	F = 55 to 500 Hz; accel = 10 G (peak); Ebb = 400 Vdc; Rp = 10,000 ohms; Ec/Ib = 10 mAdc	---	---	Ep	---	250	mVac
1042	Shock, specified pulse	13, 14	Condition A; no voltages applied	---	---	---	---	---	---
---	Torque	8, 13, 18	No voltages applied	---	---	---	---	---	---
---	Shock and torque-test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	μAdc
1002	Barometric pressure, reduced	15	Pressure = 35 mmHg (max); voltage = 2.0 kVac	---	---	---	---	---	---

NOTES:

1. The transit-time heating effect of the cathode may require compensation by a reduction in heater voltage after dynamic operation of the tubes has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.
2. The regulation or series-anode-supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.
3. The maximum instantaneous peak grid-cathode voltage shall be within the range of +175 to -750 volts.
4. Sufficient conduction, convection and forced-air cooling shall be provided to limit the envelope and anode shank temperatures to the specified maximum value under all operating conditions. Reliability will be seriously impaired if this maximum is exceeded. Where emphasis is placed on long and reliable life, lower temperatures shall be used. To insure proper cooling, the tube shall be sufficiently tightened into the heat sink (15 to 20 inch-pounds). The maximum torque, however, shall not exceed 20 inch-pounds.
5. Operation at this altitude is possible in a suitably designed circuit.
6. Sufficient conduction, convection, and forced-air cooling may be used in all electrical tests involving application of heater voltage to maintain the anode shank and seal temperatures within the specified maximum values.
7. All tests listed under quality conformance inspection, part 1, shall be performed at the conclusion of the holding period.
8. Other tube contact configurations may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the socket, jig, or cavity may be at the option of the manufacturer.
9. After heater warmup, and using the circuit of figure 2, the anode voltage shall be raised slowly to 10 kVdc. Intermittent arcing may occur as the anode voltage is increased, but more than 5 arcs shall be cause for rejection. The voltage shall be held at the 10 kVdc level for 1 minute; if any arcs occur during this time, or Ib exceeds the specified limit of 1.0 mAdc, the tube shall be rejected.

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NOTES: -Continued

10. At zero hours of life test, measure and record Cgp, and establish the drive conditions necessary to obtain a peak anode current (ib) of 3 amps (minimum) with an anode voltage of 2.0 kVdc and a bias of -40 Vdc. The pulse width of the modulator shall be 2 μs (minimum) and the duty shall be 0.0025 (maximum). At the conclusion of the life test, measure Cgp and calculate ΔCgp, and with the drive level determined at zero hours, check the peak anode current and calculate Δib.

11. Where emphasis is placed on long and reliable life, the filament voltage can safely be lowered to 6.0 volts, provided the line voltage is regulated better than $\pm$2 percent.

12. The tube shall be suitably mounted (e.g. epoxy encapsulated) and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within $\pm$20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximately logarithmic progression and shall require from 4 to 6 minutes to traverse the range. Each tube shall be vibrated for 30 minutes in the X and Y axes, except that if the cumulative result of tests on 50 or more tubes show more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need to be taken only in the position giving the higher reading. The voltages specified shall be applied during vibration. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a device having the ability to measure, with an error of less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz.

13. Testing shall be performed approximately every 90 days, with sampling as follows:

n1 = 4	c1 = 0	Where c2 represents the total allowable failures for the first and second samples combined.
n2 = 4	c2 = 1;	

In case of failure after double sampling, the failing test(s) shall become a part of quality conformance inspection (QCI), part 2, AQL 6.5, inspection level S3. After three consecutive successful submissions, the testing shall revert to the quarterly basis. None of the testing shall be considered as destructive except in case of failure.

14. The tube shall be suitably mounted (e.g. epoxy encapsulation). Each tube shall be subjected to a total of 15 shocks; 5 shocks in each plane +X, +Z, and -Z in any sequence.

15. Voltage shall be 60 Hz ac applied between the anode and grid. No other voltages shall be applied. There shall be no evidence of failure as indicated by arc-over. Perform test on 10 tubes selected from the first production lot of each year. If more than one tube fails, the test shall become part of quality conformance inspection, part 2, with an AQL of 6.5, inspection level S3. After three consecutive successful submissions, the test shall revert to a 10-tube annual test. This is not a destructive test.

16. Test shall be conducted in a cavity assembly in accordance with the following Hazeltine Corporation drawings, or equivalent:

114329-5	Final output cavity assembly.
114277-7	RF filter assembly.
114612	Connector (matching stub).
330620	Power divider (SMA"T").

A 0.47 μF cathode bypass capacitor shall be employed.

(Copies of the above drawings may be obtained from Hazeltine Corporation, Greenlawn, New York 11740 at no cost.)

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NOTES: -Continued

Ⓑ 17. This type will replace the following:

Type or part number
8538A and Y-564
ML8538 and Hazeltine P/N 910633
8933 (EIMAC)

18. The torque test shall be performed as follows:

(a) A torque of 15 inch-pounds shall be applied at right angles to the tube axis between the anode and cathode, without shock.

(b) A torque of 20 inch-pounds shall be applied at right angles to the tube axis between the anode and grid, without shock.

19. Compatibility testing is required in addition to the requirements stated herein. A sample of 10 tubes shall be subjected to compatibility testing in the Receiver Transmitter RT-868A/APX-76(V), or in RT-1026/TPX-46(V), where it shall be demonstrated that pulse train droop does not exceed 1.0 dB when used with a pulse train of 37 pulses 0.5 μs wide spaced at 2 μs. Droop shall be defined as that power difference between pulses with maximum power and minimum power respectively. There shall be no evidence of a sudden amplitude change or "step" during the pulse train.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DSA - ES

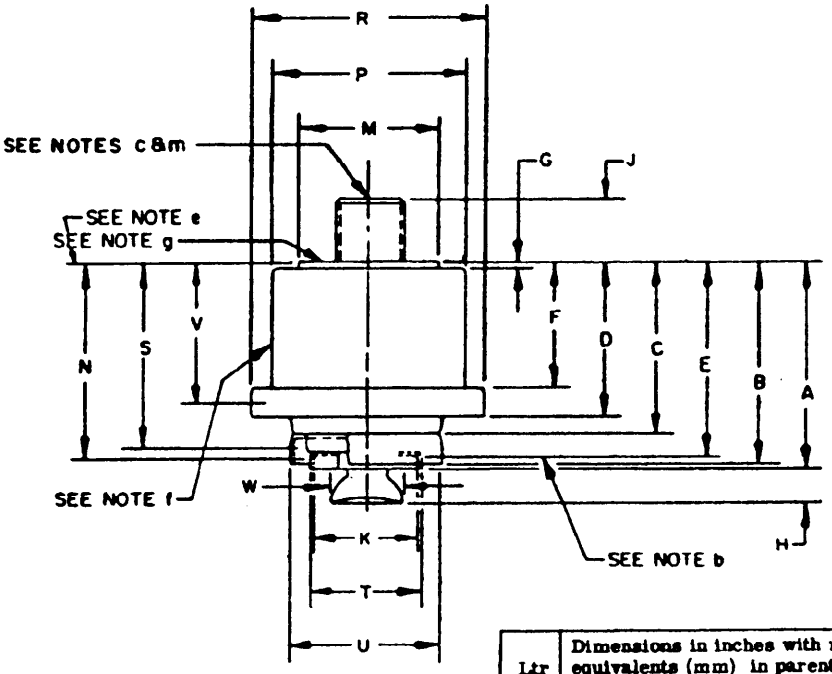
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Ⓑ Quality conformance inspection, part 2			
A	.845(21.46)	.909(23.09)	
B	.825(20.96)	.875(22.22)	
D	.640(16.26)	.669(16.09)	
F	.520(13.21)	.537(13.64)	
K	.410(10.41)	.425(10.80)	
R	.935(23.75)	.950(24.13)	a, h
T	.440(11.18)	.460(11.68)	a, d
U	.595(15.11)	.607(15.42)	a, j
Quality conformance inspection, part 3			l
C	.702(17.83)	.740(18.80)	
E	.782(19.86)	.837(21.26)	
G	.022 (.56)	.040 (1.02)	
H		.250 (6.35)	
J	.240 (6.10)	.275 (6.98)	
M	.565(14.35)	.580(14.73)	
P	.775(19.68)	.785(19.94)	
W		.313 (7.95)	
Reference dimensions			k
N	.815(20.70)		a, d, m
S	.778(19.76)		a, j, m
V	.598(15.19)		a, h, m

FIGURE 1. Outline drawing of electron tube type 8538B.

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NOTES:

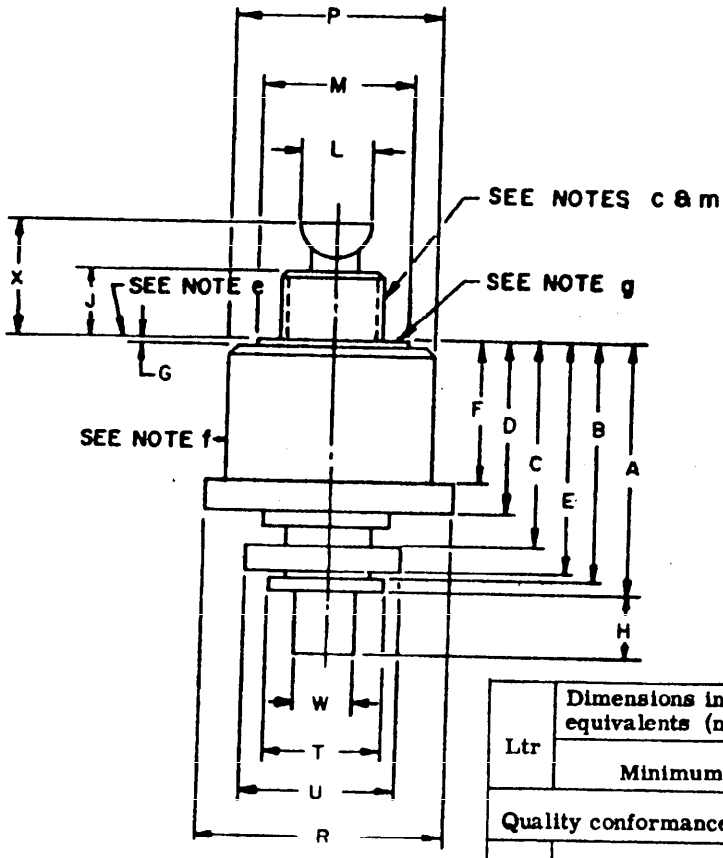
a. Total indicated runout (TIR) of contact surfaces shall be gaged from centerline of reference and shall be as follows:

Contact surface	TIR (max)	Reference
Grid	0.015	Anode
Cathode	0.015	Anode
Heater	0.015	Anode

b. See outline. Alternate heater-contact surface. Heater contact can be made to the bottom of the heater terminal-cup by means of a coil spring having a maximum coil OD of .390(9.91 mm) inch and a minimum coil ID of .320(8.13 mm) inch, or some similar device.
c. See outline. Anode, .3125-24 UNF-2A thread.
d. Heater-contact surface.
e. See outline. Reference surface. The tube shall be stopped only by this surface when screwed in the socket.
f. See outline. Insulating envelope. Do not clamp or locate on this surface.
g. See outline. Anode flange is for electrical contact; stud is for heat transfer and anode temperature measurement.
h. Grid-contact surface.
j. Cathode- or heater-contact surface.
k. Dimensions in electrode contact areas table are for socket design purposes and are not intended for inspection purposes.
l. These dimensions shall be checked on 10 tubes per month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become, for all lots in process, part of quality conformance inspection, part 2.
m. Silver plated 30 milligrams per square inch (msi) minimum plus gold strike.

FIGURE 1. Outline drawing of electron tube type 8538B - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A		.960 (24.38)	
B		.895 (22.73)	
C	.702 (17.83)	.740 (18.80)	
D	.655 (16.64)	.684 (17.37)	
E	.825 (20.96)	.875 (22.23)	
F	.537 (13.64)	.554 (14.07)	
R	.935 (23.75)	.950 (24.13)	a, h, m
T	.440 (11.18)	.460 (11.68)	a, d, m
U	.595 (15.11)	.607 (15.42)	a, j, m
W	.235 (5.97)	.265 (6.73)	a, d, m
Quality conformance inspection, part 3			1
G		.055 (1.40)	
H		.200 (5.08)	
J	.150 (3.81)	.190 (4.83)	
L		.260 (6.60)	
M	.545 (13.84)	.570 (14.48)	
P	.775 (19.68)	.785 (19.94)	
X		.340 (8.64)	

Ⓑ FIGURE 1A. Alternate outline drawing of electron tube type 8538B.

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NOTES:

a. Total indicated runout (TIR) of contact surfaces shall be gaged from centerline of reference and shall be as follows:

Contact surface	TIR (max)	Reference
Grid	0.020 (0.51)	For all TIR measurements, the reference is the anode thread pitch diameter -- see note b.
Cathode	0.020 (0.51)	
Heater	0.020 (0.51)	

b. The TIR measurement shall be made using gage per figure 3 to attach to the tube anode and reference the pitch diameter of the threads.
c. See outline. Anode, .3125-24 UNF-2A thread.
d. Heater-contact surface.
e. See outline. Reference surface. The tube shall be stopped only by this surface when screwed in the socket.
f. See outline. Insulating envelope. Do not clamp or locate on this surface.
g. See outline. Anode flange is for electrical contact; stud is for heat transfer and anode temperature measurement.
h. Grid-contact surface.
j. Cathode- or heater-contact surface.
l. These dimensions shall be checked on 10 tubes per month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause heat dimension to become, for all lots in process, part of quality conformance inspection, part 2.
m. Silver plated 30 milligrams per square inch (msi) minimum plus gold strike.

Ⓑ FIGURE 1A. Alternate outline drawing of electron tube type 8538B - Continued.

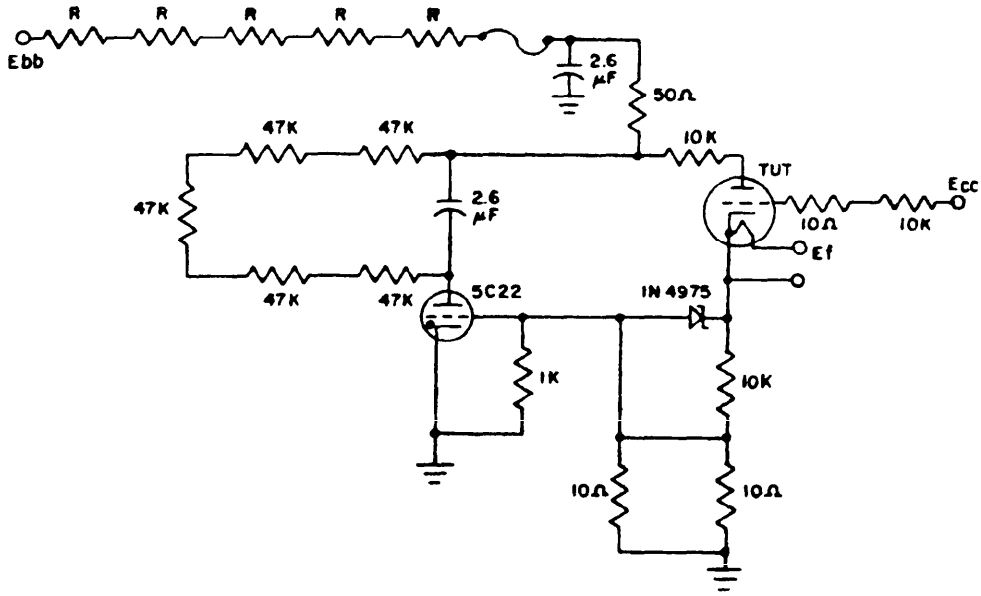
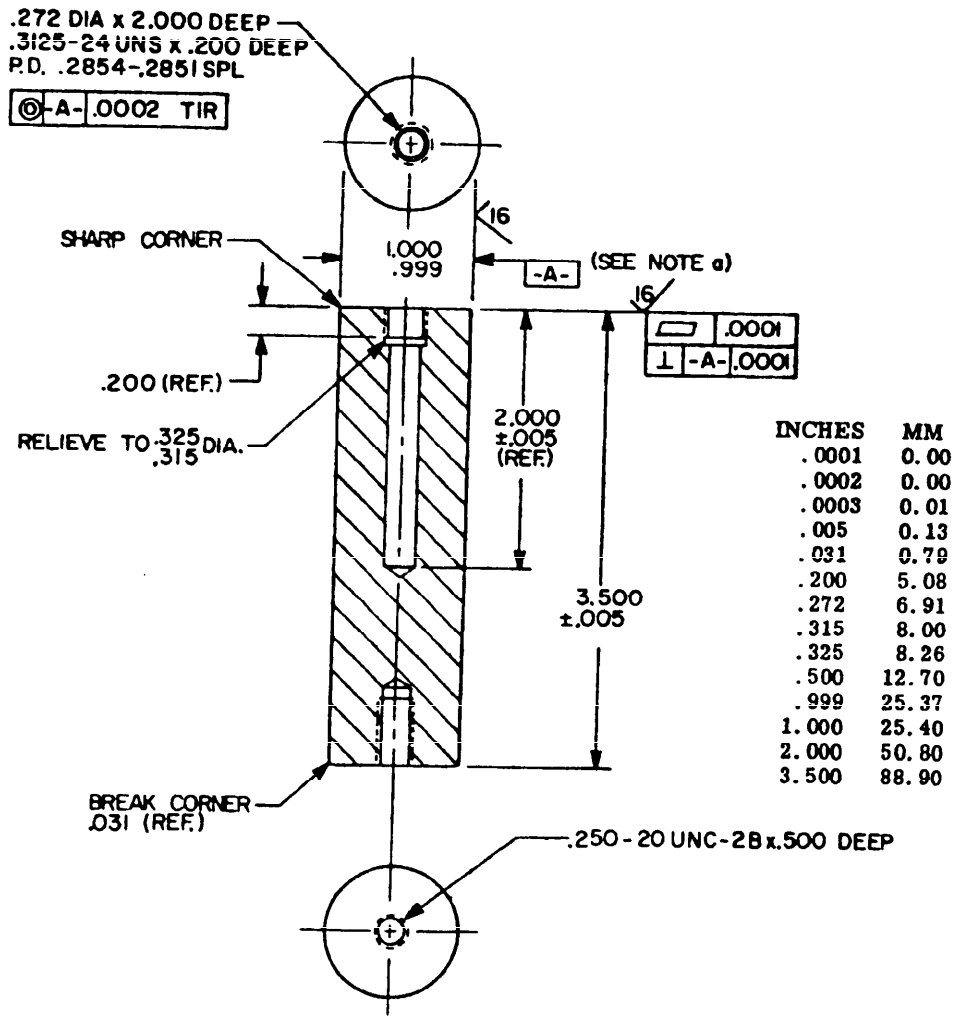


FIGURE 2. High voltage test circuit.

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INCHES	MM
.0001	0.00
.0002	0.00
.0003	0.01
.005	0.13
.031	0.79
.200	5.08
.272	6.91
.315	8.00
.325	8.26
.500	12.70
.999	25.37
1.000	25.40
2.000	50.80
3.500	88.90

NOTES:
a. Overall diameter taper .0001 maximum.
b. This arbor, or equivalent, to be used to check concentricity of grid and cathode contact surfaces with reference to anode pitch diameter (P. D.).

FIGURE 3. Check fixture arbor.

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